

MICROCANONICAL PREPARATION AND EXPONENTIAL-SCALE ENSEMBLE TRANSFER FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. The one-particle momentum-energy mark is treated as a singular paraboloid surface measure. Full-dimensional smoothing begins only in a finite compound-Poisson convolution sector whose sum map has full rank. Empty and low-particle sectors remain as explicit exponentially small terms in every high-frequency estimate. Uniform covariance and exact finite saddles are obtained from the positive compound sector. The continuous shell is centered at the extensive target and has extensive shrinking width, yielding the mixed coefficient, microcanonical pressure, constrained covariance, and prepared LDP.

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1. ROUND-SEVEN RECONSTRUCTION: SECTORWISE COMPOUND-POISSON SMOOTHING AND THE CORRECTLY SCALED SHELL

This section replaces the round-six controlling module. The one-particle momentum–energy mark is treated as a surface measure on its paraboloid. Full dimensional smoothing is obtained only after sufficiently many independent particles. Every smaller particle sector, including the empty sector, is

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retained explicitly as an exponentially small error. Normalized and extensive constraint variables are kept distinct throughout.

1.1. The singleton surface measure and its convolution rank. Let

$$C(z) = (1, v, |v|^2/2, \chi_1(z), \dots, \chi_d(z)) \in \mathbb{Z} \times \mathbb{R}^{4+d}$$

be the number and continuous constraint mark. Conditional on one particle, the momentum–energy projection is supported on $e = |p|^2/2$ and is therefore singular in four dimensions. Denote the full singleton measure by ν_1 .

For $n \geq 1$ let

$$\Phi_n(z_1, \dots, z_n) = \sum_{j=1}^n C'(z_j),$$

where C' is the continuous part. Let $D_n \Phi_n$ be its differential on the n -particle product.

Lemma 1.1 (Finite convolution rank). *There is an integer $n_0 \leq 5 + d$ and an open product set U^{n_0} of positive singleton activity on which $D_{n_0} \Phi_{n_0}$ has full rank $4 + d$. Consequently $\nu_1^{*n_0}$ has a C^4 density on an open set, and for every q there is $n(q)$ such that $\nu_1^{*n(q)}$ has a C^q density with compactly controlled derivatives on the finite saddle chart.*

Proof. Choose four velocities near $ae_1, -ae_1, be_2, -be_2$ with $a \neq b$. Variations of the four radial coordinates span the energy direction and three independent momentum directions. The remaining d marks are separated by the assumed activity patches for $\chi_1, \dots, \chi_d$. The resulting Jacobian contains a nonzero $(4 + d)$ -minor. The coarea formula gives an absolutely continuous component of the n_0 -fold convolution. Convolution with further copies differentiates under the integral and raises regularity; compactness of the saddle chart makes all bounds uniform. $\square$

1.2. Exact Poisson-sector decomposition. Let $a_{1,\varepsilon}$ be the total singleton activity and $\widehat{\nu}_{1,\varepsilon}(t, u)$ its mixed lattice–continuous Fourier transform. The singleton exponential is exactly

$$\exp\{\mu_\varepsilon a_{1,\varepsilon}(\widehat{\nu}_{1,\varepsilon}(t, u) - 1)\} = e^{-\mu_\varepsilon a_{1,\varepsilon}} \sum_{n \geq 0} \frac{(\mu_\varepsilon a_{1,\varepsilon})^n}{n!} \widehat{\nu}_{1,\varepsilon}(t, u)^n.$$

Fix $n_* = n(M)$ from Lemma 1.1. Split the sum into $n < n_*$ and $n \geq n_*$. The total probability of the first part is

$$e^{-\mu a_1} \sum_{n < n_*} \frac{(\mu a_1)^n}{n!} \leq C_{n_*} \mu^{n_*-1} e^{-c\mu}.$$

It includes the empty sector $e^{-\mu a_1}$.

Let $\mathfrak{r}_\varepsilon(t, u)$ be the connected cluster remainder. The B2-GC tree estimate gives, on the finite saddle chart,

$$\operatorname{Re} \mathfrak{r}_\varepsilon(t, u) \leq C(T + \varepsilon), \quad |\partial_u^j \mathfrak{r}_\varepsilon(t, u)| \leq C_j(T + \varepsilon)$$

for every fixed derivative order needed in the inversion. Choose the local preparation horizon so that this real-part loss is smaller than half the positive-particle singleton gap.

Theorem 1.2 (Correct mixed characteristic estimate). *For every M there are $c, C_M > 0$ such that uniformly on the exact finite saddle chart,*

$$|\varphi_\varepsilon(t, u)| \leq C_M \mu_\varepsilon^{n_*-1} e^{-c\mu_\varepsilon} + C_M (1 + \sqrt{\mu_\varepsilon}|u|)^{-M}$$

for large continuous frequency, and

$$|\varphi_\varepsilon(t, u)| \leq e^{-c\mu_\varepsilon}$$

on every compact mixed minor arc away from the lattice origin. No estimate forces the characteristic function below the empty-sector atom.

Proof. The sectors $n < n_*$ are bounded by their total Poisson probability. For $n \geq n_*$, group the first n_* marks and use the full-rank coarea density. Repeated integration by parts gives arbitrary polynomial decay after enough marks; the remaining marks have Fourier modulus at most one. The derivatives of $e^{\mu x_\varepsilon}$ are controlled by the connected bounds, and its real part is dominated by the singleton gap. On compact minor arcs, the lattice aperiodicity and the continuous convolution rank give a fixed strict modulus loss. Summing the sectors proves both estimates. $\square$

1.3. Uniform covariance and the finite saddle. Let

$$Q_\varepsilon(H, \lambda) = \mu_\varepsilon^{-1} \log \mathbb{E} \exp\{\mu_\varepsilon \langle H, X_\varepsilon \rangle + \lambda \cdot C_{\text{tot}}\}.$$

The normalized constraint mean is

$$a_\varepsilon(H, \lambda) = D_\lambda Q_\varepsilon(H, \lambda) = \mu_\varepsilon^{-1} \mathbb{E}_{H, \lambda} C_{\text{tot}}.$$

Lemma 1.3 (Compound-sector covariance). *On every compact source and multiplier chart there is $c_0 > 0$ such that*

$$D_{\lambda\lambda}^2 Q_\varepsilon(H, \lambda) \geq c_0 I$$

for all sufficiently small ε . The lower bound is supplied by the positive-activity n_0 -particle sector of Lemma 1.1; it does not assume a full-dimensional singleton density.

Proof. Condition on a small product patch inside U^{n_0} . Its Poisson probability per unit activity is bounded below uniformly on the saddle chart. The full-rank Jacobian gives a strictly positive covariance matrix on that event. Variance decomposition transfers this lower bound to the compound law. The connected correction has operator norm $O(T + \varepsilon)$ and is absorbed by choosing the preparation chart within the B2 convergence radius. $\square$

Theorem 1.4 (Exact finite-volume saddle and mean image). *Let A be a compact subset of the interior limiting mean image. For every bounded source H and $a \in A$ there is a unique exact finite-volume multiplier $\lambda_{H, a, \varepsilon}$ satisfying*

$$D_\lambda Q_\varepsilon(H, \lambda_{H, a, \varepsilon}) = a.$$

The multipliers and all finite source derivatives converge uniformly to the limiting saddle.

Proof. Lemma 1.3 gives uniform local injectivity. On the boundary of a slightly larger multiplier box, the support function of the compound mark points strictly outward because the activity patches span every constraint direction. Brouwer degree is therefore one for every target in A . Strict convexity gives uniqueness. Uniform pressure convergence and the implicit-function theorem give convergence of the saddle and its derivatives. $\square$

1.4. The correctly extensive shell. Let N_ε be the particle number and let Y_ε be the extensive continuous total. For target $a = (\rho, a')$ define

$$\mathcal{S}_\varepsilon(a) = \{N_\varepsilon = \lfloor \mu_\varepsilon \rho \rfloor, \quad Y_\varepsilon - \mu_\varepsilon a' \in \mu_\varepsilon \delta_\varepsilon B\},$$

where

$$\delta_\varepsilon \downarrow 0, \quad \sqrt{\mu_\varepsilon} \delta_\varepsilon \longrightarrow \infty.$$

Equivalently the normalized variable $Y_\varepsilon/\mu_\varepsilon$ lies in $a' + \delta_\varepsilon B$. The shell therefore contains an increasing number of central-limit standard deviations while shrinking macroscopically.

Theorem 1.5 (Mixed lattice–continuous shell coefficient). *Uniformly for a in compact subsets of the mean image and bounded sources H ,*

$$\mathbb{P}_{H, \lambda_{H,a}, \varepsilon}(\mathcal{S}_\varepsilon(a)) = \frac{1 + o(1)}{\sqrt{2\pi\mu_\varepsilon\sigma_{N|Y}^2}} \int_{\sqrt{\mu_\varepsilon}\delta_\varepsilon B} \phi_{\Sigma_{Y|N}}(y) dy + O(\mu_\varepsilon^{n_*-1} e^{-c\mu_\varepsilon}).$$

In particular the probability is subexponential and is bounded above and below by positive polynomial multiples of $\mu_\varepsilon^{-1/2}$.

Proof. Use Fourier inversion in the exact lattice coordinate and smooth upper and lower approximations of the extensive continuous shell. In the central box, expand at the exact finite saddle from Theorem 1.4; the lattice coefficient and continuous Gaussian integral give the displayed Schur complements. Compact minor arcs and the large-frequency region are controlled by Theorem 1.2. The atomic and small-particle sectors remain as the explicit exponentially small term. Boundary smoothing is $o(\mu^{-1/2})$ because $\sqrt{\mu}\delta_\varepsilon \rightarrow \infty$. $\square$

1.5. Source-dependent ensemble transfer. Define the microcanonical normalized pressure

$$Q_\varepsilon^{\text{mc},a}(H) = \mu_\varepsilon^{-1} \log \mathbb{E}^{\text{gc}} \left[e^{\mu_\varepsilon \langle H, X_\varepsilon \rangle} \mid \mathcal{S}_\varepsilon(a) \right].$$

Theorem 1.6 (Microcanonical pressure and covariance). *Locally uniformly in H ,*

$$Q_\varepsilon^{\text{mc},a}(H) \longrightarrow \inf_{\lambda} \{Q(H, \lambda) - \lambda \cdot a\} - \inf_{\lambda} \{Q(0, \lambda) - \lambda \cdot a\}.$$

Its Hessian at a regular source is the exact Schur complement

$$Q_{HH} - Q_{H\lambda} Q_{\lambda\lambda}^{-1} Q_{\lambda H}.$$

Proof. Change measure in the numerator by the exact finite saddle $\lambda_{H,a,\varepsilon}$. On the extensive shell the multiplier term is $-\mu_\varepsilon \lambda_{H,a,\varepsilon} \cdot a + o(\mu_\varepsilon)$. Theorem 1.5 makes the residual shell probability subexponential, including all atomic sectors. Repeat with $H = 0$ for the denominator. The envelope theorem and Lemma 1.3 give the Hessian. $\square$

1.6. Prepared initial large deviations. Use bounded time-zero cylinder tests and the conserved Maxwellian moment. The finite-dimensional lower bound follows from the exact saddle and pressure gap; boundary mean points are reached by mixtures of interior saddles and admissible constraint-preserving patches.

Theorem 1.7 (Prepared empirical-measure LDP). *Under the correctly scaled shell, the prepared initial empirical measure satisfies a good LDP in the weak topology augmented by the conserved Maxwellian Lyapunov moment. Its rate is the grand-canonical entropy restricted to the constraint surface and shifted by its constrained minimum.*

Proof. Theorem 1.6 gives every finite projected pressure. Exact finite-saddle tilts and strict projected covariance give local lower bounds at interior means. Convex mixtures with smooth constraint-preserving patches approximate boundary finite-rate measures. Conservation gives exponential tightness. Dawson–Gartner and the topology-upgrade lemma complete the proof. $\square$

Theorem 1.8 (Round-seven B1 closure). *The physical one-particle paraboloid is treated as a singular surface measure; smoothing begins in a full-rank compound sector; every atomic and small-sector contribution is retained; the shell is centered at the extensive total $\mu_\varepsilon a'$ with width $\mu_\varepsilon \delta_\varepsilon$; and the source-dependent coefficient, covariance, ensemble transfer, and prepared LDP follow at the exact finite saddle.*

Proof. Combine Lemmas 1.1 and 1.3 with Theorems 1.2, 1.4, 1.5, 1.6, and 1.7. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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